v1.0Particle Fusion Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Unified Nucleosynthesis Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Particle Fusion Deep Dive Request V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Fusion ignition	ICF progress noted	NIF 2023: Q>1 CONFIRMED. 3.15 MJ output from 2.05 MJ laser. FUSION_IGNITION event type added	HIGH	Historic milestone — fusion now confirmed viable; major simulation event validated
r-process site	Theory only	GW170817 kilonova: gold, platinum confirmed from NS-NS merger. MUSE 2023: Sr, Ba yields measured	HIGH	r-process site confirmed by gravitational wave observation — confirms simulation model
pp cross-section	General reference	LUNA 2022-2024: $S_{33}(0)$ = 5.12 MeV·b (3.6% improvement). S-factor lookup table updated	LOW	More precise solar fusion cross-sections improve nucleosynthesis accuracy
Gamow window	Conceptual	V2.0: Gamow peak E_G = (pi*alpha*m_r*c²/sqrt(2))$^{\{2/3\}(k\_BT)}$\{1/3\} implemented per pair	LOW	Precise tunneling calculation replaces approximate formula

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** r-process, Fusion
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Particle Fusion Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026